have had to push that maximum mass up to a hundred Suns or more. Instead, it pushed the mass down—way down, to 0.7 solar mass.

Tolman came up to Berkeley to learn more of the details. Fifty years later Volkoff recalled the scene with pleasure: “I remember being greatly overawed by having to explain to Oppenheimer and Tolman what I had done. We were sitting out on the lawn of the old faculty club at Berkeley. Amidst the nice green grass and tall trees, here were these two venerated gentlemen and here was I, a graduate student just completing my Ph.D., explaining my calculations.”

Now that they knew the masses of neutron stars in an idealized universe with no nuclear force, Oppenheimer and Volkoff were ready to estimate the influence of the nuclear force. Here the formulas that Tolman had worked out so carefully for various hypothetical equations of state were helpful. From Tolman’s formulas one could see roughly how the star’s structure would change if the nuclear force was repulsive and thereby made the equation of state more “stiff” than the one Volkoff had used, and the change if it was attractive and thereby made the equation of state more “soft.” Within the range of believable nuclear forces, those changes were not great. There must still be a maximum mass for neutron stars, Tolman, Oppenheimer, and Volkoff concluded, and it must lie somewhere between about a half solar mass and several solar masses.

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*See the discussion in the last section of Chapter 1 (“The Nature of Physical Law”) of the relationship between different descriptions of the laws of physics and their domains of validity.

Oppenheimer and Volkoffs conclusion cannot have been pleasing to people like Eddington and Einstein, who found black holes anathema. If Chandrasekhar was to be believed (as, in 1938, most astronomers were coming to understand he should), and if Oppenheimer and Volkoff were to be believed (and it was hard to refute them), then neither the white-dwarf graveyard nor the neutron-star graveyard could inter massive stars. Was there any conceivable way at all, then, for massive stars to avoid a black-hole death? Yes; two ways.

First, all massive stars might eject so much matter as they age (for example, by blowing strong winds off their surfaces or by nuclear explosions) that they reduce themselves below 1.4 solar masses and enter